WORKED SOLUTION

STEP II 2021, Question 12

Probability and Statistics • Gambler's Ruin

STEP II 2021, Question 12

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Each match keeps returning to a position it has been in before. So we don't add up an infinite list of ways to win. We let P be the probability of winning from that position, condition on what happens next, and get an equation in which P appears on both sides. The same trick, applied to the cautious game, turns it into the match of part (ii).

Solution

Part (i)

Let P be the probability that A wins the match. Condition on the first game. A wins it with probability p_{A} and the match is over. B wins it with probability p_{B} and A has lost. Otherwise the game is drawn, with probability $1 - p_{\mathrm{A}} - p_{\mathrm{B}}$, and since games are independent the match starts afresh, so A's chance from there is again P. Hence

$$P = p_{\mathrm{A}} + (1 - p_{\mathrm{A}} - p_{\mathrm{B}})P \quad \Longrightarrow \quad (p_{\mathrm{A}} + p_{\mathrm{B}})P = p_{\mathrm{A}} \quad \Longrightarrow \quad P = \frac{p_{\mathrm{A}}}{p_{\mathrm{A}} + p_{\mathrm{B}}},$$

dividing by $p_{\mathrm{A}} + p_{\mathrm{B}} > 0$. (Equivalently, A wins at game k with probability $(1 - p_{\mathrm{A}} - p_{\mathrm{B}})^{k-1} p_{\mathrm{A}}$, and the geometric series converges because $0 < p_{\mathrm{A}} + p_{\mathrm{B}} < 1$.)

Part (ii)

Let D be A's number of wins minus B's after some games. Each game changes D by $+1$ or -1, and the match ends the first time $|D| = 2$.

Even length. After n games, with a wins for A and b for B, we have $a + b = n$ and $D = a - b = n - 2b$, so D has the same parity as n. The match ends when $D = \pm 2$, which is even, so n is even.

Probability. Look at the games in pairs. The match is level ($D = 0$) at the start, and since it can only end after an even number of games, it is level again after every pair that doesn't finish it. From a level position, the next two games are AA with probability

p^2 (A wins), BB with probability q^2 (B wins), or AB or BA with total probability $2pq$, which leaves the match level again. Let P be the probability that A wins from a level position. Then

$$P = p^2 + 2pq\,P \quad \Longrightarrow \quad P = \frac{p^2}{1-2pq}.$$

Since $p+q=1$, $1=(p+q)^2 = p^2+2pq+q^2$, so $1-2pq = p^2+q^2$ and

$$P = \frac{p^2}{p^2+q^2},$$

as required. Here $p^2+q^2>0$, so the division is fine.

Part (iii)

Cautious version. The number of tokens goes up or down by one each round, starting at 1 and stopping at 0 (loss) or 4 (win). It can't jump past 4, so the player wins exactly when the count reaches 4 before 0.

> **Ask yourself** From 2 tokens, what does the player need? Have we met this before?

From 2 tokens the player wins by getting two more round wins than losses (reaching 4) before getting two more losses than wins (reaching 0). That is the match of part (ii), with the player as A and the game of chance as B. So from 2 tokens the probability of winning is $\dfrac{p^2}{p^2+q^2}$.

From 1 token, the first round must be won, or the player drops to 0 and loses. So the first round takes the player to 2 tokens with probability p, and the rounds are independent, so

$$P_{\text{cautious}} = p \cdot \frac{p^2}{p^2+q^2} = \frac{p^3}{p^2+q^2}.$$

(If you prefer equations: with P_k the probability of winning from k tokens, $P_1 = pP_2$, $P_2 = pP_3 + qP_1$ and $P_3 = p + qP_2$ give the same answer.)

Bold version. From 1 token the count goes to 2 or 0, and from 2 it goes to 4 or 0. So the player wins only by winning the first two rounds:

$$P_{\text{bold}} = p^2.$$

Comparison. Both probabilities are positive, so we compare their ratio with 1:

$$\frac{P_{\text{cautious}}}{P_{\text{bold}}} = \frac{p}{p^2+q^2}.$$

Since $p^2+q^2>0$, this ratio is greater than 1 if and only if

$$p > p^2+q^2 \iff p-p^2 > q^2 \iff pq > q^2 \iff p > q,$$

dividing by $q > 0$ at the last step (valid because $p < 1$). And $p > q = 1 - p$ exactly when $p > \frac{1}{2}$. In the same way, the ratio is less than 1 exactly when $p < q$, that is, when $p < \frac{1}{2}$. So the cautious version is better when $\frac{1}{2} < p < 1$ and the bold version is better when $0 < p < \frac{1}{2}$. (At $p = \frac{1}{2}$ both give $\frac{1}{4}$.)

Checks

- At $p = \frac{1}{2}$ the cautious game is a fair random walk, and the chance of reaching 4 before 0 from 1 is $\frac{1}{4}$. Our formula gives $\frac{1}{8}/\frac{1}{2} = \frac{1}{4}$, matching $p^2 = \frac{1}{4}$ for the bold game.
- At $p = 0.7$: cautious gives $0.343/0.58 \approx 0.591$ and bold gives 0.49. At $p = 0.3$: cautious gives $0.027/0.58 \approx 0.0466$ and bold gives 0.09. Both agree with a computer check that follows the token distribution round by round.
- In part (ii), $p = \frac{1}{2}$ gives $\frac{1}{2}$, as symmetry demands.

Common slips

- In part (ii), treating "AB" and "BA" as one outcome of probability pq, rather than $2pq$.
- Forgetting to justify that a non-final pair of games always returns the match to level. It does, because the match can't end after an odd number of games.
- In the comparison, multiplying out or dividing without saying the quantities are positive.

Beyond STEP

The cautious game is the *gambler's ruin* problem. It is a random walk with absorbing barriers, and the general answer for starting at k with target N is $\dfrac{1-(q/p)^k}{1-(q/p)^N}$ when $p \neq q$. The result that bold play is best in an unfavourable game, and timid play in a favourable one, goes back to Dubins and Savage's study of optimal gambling. Grimmett and Welsh, *Probability: An Introduction*, treats random walks and gambler's ruin through exactly this kind of first-step conditioning.

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